A. IoT Fundamentals

1. What is IoT?
IoT is a network of physical devices connected to the internet to collect and exchange data.

2. What are the four layers of IoT architecture?
Perception, Network, Processing and Application layers.

3. What is a sensor?
A sensor detects physical changes such as temperature, light, humidity or pressure.

4. What is an actuator?
An actuator converts electrical signals into physical actions, such as motor movement.

5. What is an embedded system?
A computer system designed to perform a specific dedicated task.

6. What is the difference between IoT and embedded systems?
Embedded systems perform dedicated tasks, while IoT devices additionally communicate and exchange data over networks.

B. Arduino and Programming

7. What is Arduino Uno?
An open-source microcontroller development board based on ATmega328P.

8. What is a microcontroller?
A compact integrated circuit containing a processor, memory and input/output peripherals.

9. What is the difference between setup() and loop()?
setup() executes once, whereas loop() executes repeatedly.

10. What is pinMode()?
It configures an Arduino pin as INPUT, OUTPUT or INPUT_PULLUP.

11. What is the difference between digitalRead() and analogRead()?
digitalRead() reads HIGH/LOW, while analogRead() reads analog values from 0–1023 on Arduino Uno.

12. What is PWM?
Pulse Width Modulation controls power by varying the duty cycle of digital pulses.

13. What is ADC?
Analog-to-Digital Converter converts analog signals into digital values.

C. Sensors and Components

14. What is an ultrasonic sensor?
It measures distance using ultrasonic sound waves.

15. What is the working principle of HC-SR04?
It sends ultrasonic waves and measures the time taken for the reflected waves to return.

16. What is LDR?
Light Dependent Resistor changes its resistance according to light intensity.

17. What is MQ-2 sensor?
A gas sensor used to detect gases such as LPG, methane and smoke.

18. What is a relay?
An electrically operated switch used to control high-power devices using low-power signals.

19. What is an H-Bridge?
An electronic circuit used to control the direction of a DC motor.

20. What is a pH sensor?
A sensor used to measure the acidity or alkalinity of a substance.

21. What is the difference between analog and digital sensors?
Analog sensors produce continuous signals, while digital sensors provide discrete digital outputs.

D. Communication Protocols

22. What is UART?
Universal Asynchronous Receiver Transmitter is a serial communication method using TX and RX lines.

23. What is baud rate?
It is the number of signal symbols transmitted per second; Arduino examples commonly use 9600 baud.

24. What is I2C?
Inter-Integrated Circuit is a communication protocol using two lines: SDA and SCL.

25. What is SPI?
Serial Peripheral Interface is a synchronous communication protocol commonly using MOSI, MISO, SCK and SS.

26. What is MQTT?
Message Queuing Telemetry Transport is a lightweight messaging protocol used in IoT.

E. General Lab and Practical Questions

27. What is Tinkercad?
An online platform used to design and simulate electronic circuits.

28. What is the purpose of Serial Monitor?
It displays data exchanged between Arduino and the computer.

29. What is the difference between delay() and delayMicroseconds()?
delay() works in milliseconds, while delayMicroseconds() works in microseconds.

30. What is the difference between HIGH and LOW?
HIGH represents a logic-high voltage and LOW represents a logic-low voltage.